

Leadani LFS

DigitalOcean Migration — Technical Brief

Prepared for Developer / DevOps

1. Stack Overview

- Single Node.js / Express server — serves REST API and React frontend (SPA) as static files
- Frontend built with Vite !' output lands in dist/public/; served by Express in production
- Build command: npm run build • Start command: npm start
- Runtime: Node.js 20+ | Package manager: npm
- Real-time: Socket.io (runs on same HTTP port — no separate WebSocket server needed)
- ORM: Drizzle ORM with PostgreSQL | Auth: JWT (stateless, no Redis session store needed)

2. Compute — Recommended DO Setup

Option	Notes
App Platform Basic	Simplest — DO manages deploys, SSL, and scaling
2 GB Droplet (1 vCPU)	More control; needs Nginx + Certbot for HTTPS

- Single process is fine for a start — no clustering required
- If using multiple replicas behind a load balancer: enable sticky sessions for Socket.io

3. Database Migration

1. pg_restore into DO Managed Postgres
 2. Update DATABASE_URL env var to point at DO Postgres connection string
 3. Current DB driver: @neondatabase/serverless (Neon-specific WebSocket transport)
- Required change: swap to standard pg driver + drizzle-orm/node-postgres
 - Schema, queries, and Drizzle models remain 100% unchanged — driver-only swap

- Pool size: currently max 30 connections — tune to DO plan connection limit

4. Required Environment Variables

Variable	Purpose
DATABASE_URL	PostgreSQL connection string (postgresql://user:pass@host:5432/db)
JWT_SECRET	Strong random string for signing JWT auth tokens
HMAC_SECRET	Secret for validating incoming webhook HMAC signatures
PORT	Listen port — App Platform sets this automatically
NODE_ENV	Must be set to production
FRONTEND_URL	Full production URL — controls Socket.io CORS allowed origin
VAPID_PUBLIC_KEY	Web Push public key (generate once with npx web-push generate-vapid-keys)
VAPID_PRIVATE_KEY	Web Push private key
SARVAM_API_KEY	Sarvam AI — Quality Check, AI Lead Rating, Insta Support
WAUPER_API_KEY	Wauper Session Messaging API — Saila.AI WhatsApp engine

Optional (Google Sheets backup): GOOGLE_SERVICE_ACCOUNT_EMAIL • GOOGLE_PRIVATE_KEY
Meta WhatsApp credentials are stored per-company in the database — not in env vars.

5. Socket.io / Real-time

- Runs on the same HTTP port as Express — no separate WebSocket server or port
- Authentication: JWT token passed via `socket.handshake.auth.token` (validated on connect)
- Rooms used: `user:{id}`, `company:{id}`, `company_admins:{id}`, `sheet:{id}`
- If using a load balancer / reverse proxy:
 - Enable WebSocket upgrade header pass-through (`proxy_pass` with Upgrade headers in Nginx)
 - Enable sticky sessions (`ip_hash` or consistent hashing) so socket connections stay on same instance
- Alternative: deploy a single replica to avoid sticky session complexity

6. Web Push / PWA

- Used for push notifications (attendance alerts, etc.)
- Generate VAPID keys ONCE: `npm run web-push generate-vapid-keys`
- Store `VAPID_PUBLIC_KEY` and `VAPID_PRIVATE_KEY` permanently as env vars

WARNING: Changing VAPID keys breaks ALL existing push subscriptions. Generate once and never rotate.

7. External Services to Keep

Service	Credential	Used for
Sarvam AI	SARVAM_API_KEY	Quality Check, AI Lead Rating, Insta Support chat
Wauper API	WAUPER_API_KEY	Saila.AI WhatsApp message sending
Meta WhatsApp Business	Stored in DB (per-company)	WhatsApp Cloud API webhook receiver
Google Sheets API	Service account credentials	Hourly automatic lead data backup

- All external API calls are made server-side — no API keys are exposed to the browser

8. Developer Checklist

- 1 Swap DB driver: `@neondatabase/serverless` !' `pg` + `drizzle-orm/node-postgres` (in `server/db.ts`)
- 2 Export Neon DB using `pg_dump` !' `pg_restore` into DO Managed Postgres instance
- 3 Set all required environment variables (Section 4) in DO App Platform or Droplet
- 4 Run `npm run db:push` pointing at the new DO Postgres DB to verify schema sync
- 5 Set `FRONTEND_URL` to the production domain (critical for Socket.io CORS)
- 6 Configure reverse proxy (Nginx / App Platform) to pass WebSocket upgrade headers